

Azhar Mohammed
DevOps Engineer

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PROFESSIONAL SUMMARY

- DevOps Engineer with 5+ years of experience in AWS and Azure, specializing in CI/CD, Kubernetes, Terraform, and Cloud Security (HIPAA, SOC2). Terraform Enterprise, Packer, GitHub Enterprise (SAML/SSO), JFrog Artifactory, Nginx ingress controllers, disconnected environment deployments.
- Skilled in infrastructure automation, observability, and compliance-driven DevOps. Experienced across roles titled DevOps Engineer, Cloud Security Engineer, Site Reliability Engineer (SRE), and Platform Engineer Proven expertise in implementing **CI/CD pipelines**, managing **containerized workloads with security guardrails**, and leveraging IaC to ensure **scalable, secure, and auditable infrastructure**. Skilled in building **security automation pipelines** with Python/Go for compliance checks, incident response, and cloud posture management.
- Experienced in GITHUB/Gerrit workflows, Git flow, Python-based tooling, Embedded Linux, and Yocto build system.
- Architected and modularised cloud-native infrastructure using **Terraform, Azure CLI**, and **AWS CloudFormation**, provisioning **AKS, EKS, Function Apps, Kafka**, and **IAM roles** with **YAML-driven policy-as-code** and **remote state isolation**.
- Designed and implemented **Jenkins pipelines (Groovy DSL)** and **GitHub Actions workflows** for multi-stage deployment of **Helm-based microservices**, integrating **Vault secret injection**, **Docker image scanning**, and **Terraform plan validation**.
- Managed production **Kubernetes clusters (AKS, EKS, Minikube)** with **RBAC**, **pod security policies**, **taints/tolerations**, and **affinity rules**, ensuring compliant and fault-tolerant workloads across dev, staging, and prod environments.
- Orchestrated containerised platforms using **Docker, Docker Compose**, and **Docker Swarm**, integrating **multi-host networking**, **image caching strategies**, and **label-based HA constraints** for ephemeral services.
- Developed and integrated **Ansible playbooks** for provisioning **Linux VMs**, configuring **systemd services**, and hardening **Ubuntu, Amazon Linux**, and **CentOS** nodes via **cloud-init** and **role-based YAML templates**.
- Embedded **HashiCorp Vault** into CI/CD pipelines and IaC workflows, managing **AppRole token provisioning**, **ephemeral PKI certs**, and **dynamic Azure secrets** with **TTL-based credential rotation**.
- Automated topic creation, ACL policies, and schema enforcement in **Apache Kafka** clusters using **Terraform, Airflow DAGs**, and **Python clients**, integrated with **Zookeeper** for controller reconciliation.
- Customized **Prometheus exporters** using **Python** to expose **Kafka lag**, **Airflow DAG metrics**, and **AKS pod health**, visualized through **multi-tenant Grafana dashboards** and tied to actionable alert rules.
- Deployed centralized observability stacks using **ELK (Filebeat, Logstash, Kibana)** and **Azure Monitor**, parsing structured logs from **Jenkins, Kubernetes**, and **Function Apps**, with filter-based routing to **Log Analytics**.
- Automated **AWS Lambda** and **Azure Functions** deployments for event-driven tasks such as **tag compliance**, **EC2 snapshot rotation**, and **AKS node patch validation**, integrating triggers from **CloudWatch** and **Event Grid**.
- Developed compliance-driven automation scripts using **Python, PowerShell**, and **Bash**, validating **S3 lifecycle policies**, **IAM anomalies**, and **AKS resource governance** across cloud tenants.
- Integrated **Trivy** scanning into **Dockerfile pipelines** and enforced CVE remediation gates using **Jenkinsfile** stages, with Slack-based approvals for bypasses and inline documentation for exception handling.
- Troubleshoot distributed microservices via **kubectl logs**, **Helm diff**, and **Grafana/ELK traces**, identifying failures such as **crash loops**, **OOM kills**, and **container image pull errors** in live clusters.
- Enabled **Airflow DAG validation**, **retry orchestration**, and downstream sync with **Kafka** using **custom Python sensors**, tied to **Event Grid** for failure notifications and conditional branching.
- Coordinated with **SRE teams** to define and align **SLIs/SLOs** across **Grafana, Prometheus**, and **Logstash**, ensuring consistent **metrics instrumentation**, **log enrichment**, and **dashboard ownership**.
- Maintained versioned infrastructure standards with **GitOps workflows**, GitHub branch protection, and codeowners, embedding CI hooks for **Terraform validate**, **Ansible lint**, and **Dockerfile best practices**.

TECHNICAL SKILLS

1. **Cloud Platforms & Services:** AWS (EC2, S3, IAM, Lambda, CloudFormation, VPC, CloudWatch, SNS), Azure (AKS, Azure Functions, Azure CLI, Azure Monitor, Log Analytics, Event Grid), Docker Swarm, Kubernetes (AKS, EKS, Minikube)
2. **Infrastructure as Code:** Terraform Enterprise, Packer, AWS CloudFormation, Azure CLI, cloud-init, YAML
3. **Automation & Configuration Management:** Ansible, Bash, Python, PowerShell, GitHub Actions
4. **CI/CD & Build Systems:** Jenkins, GitHub, GitHub Enterprise (SAML/SSO), JFrog Artifactory, Jenkinsfile, Slack Integrations
5. **Containerization & Orchestration:** Docker, Docker Compose, Kubernetes, Helm, Nginx ingress controllers.
6. **Monitoring, Logging & Observability:** Prometheus, Grafana, ELK Stack, Nagios, Azure Monitor, Log Analytics
7. **Secrets & Security:** HashiCorp Vault, IAM policy validation, certificate automation
8. **Data Streaming & Workflow Orchestration:** Apache Kafka, Apache Airflow, Zookeeper, DAG orchestration
9. **Scripting & Programming:** Python, Bash, PowerShell
10. **CI/CD & Build Systems** → Jenkins (DSL, Pipelines), GitHub, GitHub Actions, **Gerrit, Git Flow, Yocto**
11. **Operating Systems** → Linux (Ubuntu, CentOS, Amazon Linux), **Embedded Linux**

WORK EXPERIENCE

RingCentral – San Francisco, United States

Jan 2025 – Present

DevOps Engineer

- Engineered modular **Terraform** configurations for provisioning **Azure AKS**, **Azure Functions**, **Kafka brokers**, and **Managed Identities**, embedding policy-as-code and state isolation across environments.
- Implemented GitHub-triggered **Jenkins pipelines (DSL)** for deploying **Helm-based services** to **Kubernetes**, integrating **Vault secrets injection**, **Slack status hooks**, and **Terraform plan validations**.
- Provisioned infrastructure with **Terraform Enterprise** (workspaces, remote state, policy checks).
- Automated **Packer-based AMI builds** integrated into Jenkins pipelines.
- Deployed workloads with **Nginx ingress controllers** for ingress routing and TLS termination.
- Managed Docker images and Helm charts via **JFrog Artifactory** for artifact lifecycle.
- Delivered infrastructure in **disconnected/air-gapped environments** for compliance-driven deployments.
- Supported development teams by troubleshooting **GitHub/Gerrit** workflow issues and maintaining Git Flow strategies.
- Developed/refactored Python tools to automate build validation and release workflows in Embedded Linux.
- Documented CI/CD processes, branching strategies, and troubleshooting guides to support development teams.
- Managed **production Kubernetes clusters (AKS)** by enforcing **namespace RBAC**, **pod security contexts**, **taints/tolerations**, and **liveness/readiness probes** for high-availability workloads.
- Refactored infrastructure lifecycle workflows into consolidated **Ansible** playbooks, automating **Vault onboarding**, **cloud-init templating**, and **Linux hardening** using **YAML-driven roles**.
- Integrated **HashiCorp Vault** with **Ansible**, **Jenkins**, and **Python**, provisioning **AppRole** tokens, rotating **dynamic Azure credentials**, and issuing ephemeral **PKI certificates** with enforced TTLs.
- Orchestrated **Kafka topic automation** using **Airflow DAGs**, **Python scripts**, and **Terraform bindings**, coordinating with **Zookeeper configs** and AKS-hosted ingestion consumers.
- Created custom **Prometheus exporters** in **Python** for collecting **Kafka consumer lag**, **Airflow DAG durations**, and **container metrics**, integrated into multi-tenant **Grafana dashboards** with alert rules.
- Tuned **Azure Monitor** and **Log Analytics queries** to visualize **Kubernetes pod memory leaks**, **Function App errors**, and **AKS node saturation**, routing critical alerts to DevOps channels via webhooks.
- Deployed ELK pipelines using **Filebeat**, **Logstash**, and **Kibana**, normalizing **AKS pod logs**, **Function traces**, and **Jenkins builds** into searchable indices with retention filters and structured fields.
- Managed **Docker Swarm** workloads with **label-based constraints**, **multi-host networking**, and **HA failover rules**, supporting ephemeral analytics jobs alongside long-running services.
- Automated provisioning and validation of **Kafka clusters** using **Terraform**, **Ansible**, and **Jenkins**, embedding TLS encryption, Zookeeper integration, and topic policy enforcement.
- Used **PowerShell** and **Azure CLI** to automate patching, inventory audits, and compliance validation on **Function Apps**, **AKS nodes**, and **Linux VMs**, integrating outputs into Slack via webhooks.
- Embedded **Vault PKI** with internal service mesh by generating **short-lived TLS certs**, binding them to **Helm deployments**, and validating trust chains through automated Ansible hooks.
- Triggered **Airflow tasks** for DAG state validation, downstream Kafka readiness, and error propagation using **custom Python sensors**, integrating with **Azure Event Grid** for retry orchestration.
- Collaborated with SRE and platform teams to align **Grafana dashboards**, **Prometheus metrics**, and **Logstash filters** with SLO definitions, surfacing performance issues across distributed microservices.

Environment: Terraform, Azure, Kubernetes, Jenkins, Apache Kafka, Apache Airflow, Docker Swarm, Docker, Helm, GitHub, Ansible, HashiCorp Vault, Prometheus, Grafana, ELK Stack, Python, PowerShell, Bash, Linux, cloud-init, systemd, YAML, Slack Integrations

DigitalOcean – San Francisco, California, US

Oct 2023 – Dec 2024

DevOps Engineer – AWS

- Developed end-to-end **Jenkins pipelines** using **Groovy DSL**, integrating **Docker builds**, **unit testing**, **AWS CLI deployments**, and Slack-based status notifications for containerized services.
- Resolved **GitHub/Gerrit CI/CD** pipeline issues and supported developers with Git Flow and release workflows.
- Authored reusable **Terraform modules** for provisioning **AWS EC2 instances**, **S3 buckets**, and **IAM roles**, aligning infrastructure with environment-specific parameterisation and policy compliance.
- Optimized AWS networking using **Prefix Lists**, **quota management**, and **VPC peering strategies**.
- Integrated **GitHub Enterprise with SAML/SSO** for secure CI/CD automation.
- Orchestrated containerized microservices using **Docker** and local **Minikube clusters**, leveraging **kubectrl**, **Helm charts**, and manifests to test pod health, network policies, and application ingress rules.
- Maintained and enhanced **AWS CloudFormation stacks** for provisioning **VPCs**, **route tables**, **NAT gateways**, and **security groups**, standardizing multi-region network setups for microservices.
- Integrated lightweight **AWS Lambda** functions with **CloudWatch Events** and **SNS**, enabling dynamic resource tagging, compliance validation, and reactive automation for EC2 and S3 activities.
- Automated AWS resource validation using **Python (Boto3)** scripts to scan **ELB health**, **S3 inventory**, and IAM policy mismatches, embedding checks into CI environments for drift detection.

- Installed and configured **Prometheus node exporters** to collect system metrics from Docker hosts, with custom **Grafana dashboards** for visualizing resource utilization across services and clusters.
- Implemented centralized logging using **Filebeat** and **Logstash**, shipping logs from **Docker containers** and host OS into the **ELK Stack**, supporting traceability for CI/CD and runtime issues.
- Integrated **CloudFormation** workflows into **Jenkins pipelines**, using **AWS CLI**, secure credential injection, and PR-triggered dynamic environment provisioning for dev/test branches.
- Created hardened **Docker images** with custom Dockerfile definitions, applying **multi-stage builds**, **alpine base layers**, and **trivy-based vulnerability scanning** in the CI cycle.
- Tuned **Ansible inventory structures** using dynamic host groups and roles to manage Linux server state across environments, ensuring consistent provisioning for integration and regression pipelines.
- Wrote **AWS CLI** and **Python scripts** to automate lifecycle policies and remove unused **S3 objects**, deregister **old AMIs**, and cleanup **EBS snapshots** from legacy deployments.
- Managed Docker Compose-based local environments for multi-container microservices, defining interdependent service layers, health checks, and shared volumes for reproducibility.
- Defined **Grafana alerting rules** tied to **Prometheus thresholds**, enabling actionable alerts on pod memory spikes, node saturation, and system downtime without third-party dependencies.

Environment: AWS (EC2, S3, IAM, Lambda, CloudFormation, CloudWatch, SNS, VPC), Terraform, Ansible, Jenkins (Groovy DSL), GitHub, GitHub Actions, Docker, Kubernetes, Minikube, Helm, Python (Boto3), Bash, Prometheus, Grafana, ELK Stack, Filebeat, Trivy, Linux (Amazon Linux 2, Ubuntu)

Mphasis – Bengaluru, India

June 2021 – Sep 2023

Junior DevOps Engineer – AWS

- Authored and maintained **AWS CloudFormation** templates to define and provision **EC2 instances**, **S3 buckets**, **IAM roles**, and basic **VPC configurations**, establishing repeatable and version-controlled infrastructure provisioning processes.
- Developed and orchestrated **Jenkins freestyle jobs** to automate build, test, and deployment stages for backend services, incorporating GitHub webhook triggers, artifact archiving, and custom post-build actions.
- Created modular **Bash scripts** to automate infrastructure tasks such as log backups to **S3**, system health checks, EBS volume snapshots, and scheduled maintenance operations across **Ubuntu** and **CentOS** environments.
- Built and deployed lightweight application containers using **Docker**, designing custom Dockerfile instructions, environment variable management, and multi-stage builds to optimize image size and runtime efficiency.
- Provisioned and validated **Kubernetes workloads** on **Minikube**, managing YAML configurations for pods, deployments, and services, and troubleshooting container states using kubectl, logs, and readiness/liveness probes.
- Integrated **GitHub repositories** with Jenkins pipelines for CI enablement, configuring repository webhooks, managing branch strategies, and enforcing automated test/lint checks at merge points.
- Configured and customized **Nagios monitoring**, developing basic plugins in **Bash** for tracking CPU usage, memory thresholds, disk capacity, and service availability across multiple cloud-hosted nodes.
- Supported deployment and testing of **AWS Lambda** functions for event-driven automation tasks such as security group compliance validation and scheduled EC2 inventory reporting.
- Wrote and applied basic **Ansible playbooks** to automate configuration tasks, including package installation, service enablement, and hardening steps for cloud-hosted Linux instances.
- Utilized **Python** scripts to interact with **AWS SDK (Boto3)** for managing S3 object metadata, generating IAM user reports, and automating tagging validations on cloud resources.
- Maintained CI/CD documentation, Docker image versioning standards, and infrastructure onboarding guides to ensure consistency and cross-team adoption of DevOps practices.
- Collaborated closely with developers, testers, and cloud architects to troubleshoot CI failures, container startup issues, and infrastructure configuration drifts, reinforcing team alignment on operational readiness.

Environment: AWS (EC2, S3, IAM, Lambda, CloudFormation, VPC), Jenkins, GitHub, Docker, Docker Compose, Kubernetes (Minikube), Nagios, Bash, Python, YAML, Linux (Ubuntu, CentOS)

CitiusTech – Mumbai, India

Nov 2020 – May 2021

Cloud Operations Intern

- Monitored EC2 and Lambda metrics via **CloudWatch**, and automated daily health checks using **Python** scripts and **AWS CLI**.
- Wrote **Bash scripts** to automate log rotation, system updates, and service monitoring across **Linux EC2** instances.
- Contributed to **Terraform** modules to provision **IAM roles**, **S3 buckets**, and security groups in dev environments.
- Built and pushed **Docker images** to a private registry; debugged container issues using logs and docker exec.
- Deployed and validated **Kubernetes workloads** (pods, deployments, services) on **EKS**, including readiness probes.
- Created **GitHub Actions** pipelines for internal Python apps to automate testing, linting, and build artifacts.
- Maintained **Jenkins pipelines** with health checks and notifications for staging deployments.
- Built **Grafana dashboards** with **Prometheus** metrics to track CPU/memory usage of dev containers.
- Documented automation workflows and participated in **daily stand-ups** and **sprint retrospectives** using Agile tools.

Environment: AWS (EC2, S3, IAM, Lambda, CloudWatch), Terraform, Jenkins, GitHub Actions, Docker, Kubernetes (EKS), Bash, Python, Prometheus, Grafana, Linux (Ubuntu)

EDUCATION

Master's: Management in Data Analytics, Indian Wesleyan University, Marion, IN.

Bachelor's: Information Technology, Jawaharlal Nehru Technological University, Hyderabad, India.